

Test your knowledge of CART and Random Forests

1. In this problem, you will fit regression trees to the **Boston** dataset. This data was part of an important paper in 1978 by Harrison and Rubinfeld titled - “Hedonic housing prices and the demand for clean air” published in the Journal of Environmental Economics and Management 5(1): 81-102. The dataset has the following fields:

- **crim**: per capita crime rate by town
- **zn**: proportion of residential land zoned for lots over 25,000 sq.ft
- **indus**: proportion of non-retail business acres per town
- **chas**: Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
- **nox**: nitrogen oxides concentration (parts per 10 million)
- **rm**: average number of rooms per dwelling
- **age**: proportion of owner-occupied units built prior to 1940
- **dis**: weighted mean of distances to five Boston employment centres
- **rad**: index of accessibility to radial highways
- **tax**: full-value property-tax rate per \$10,000
- **prratio**: pupil-teacher ratio by town
- **black**: $1000(Bk - 0.63)^2$ where Bk is the proportion of blacks by town
- **lstat**: lower status of the population (percent)
- **medv**: median value of owner-occupied homes in \$1000s

We will try to predict the median house value using thirteen predictors

- (a) Use a seed of 1. Split the dataset into a training set and a test set using the **sample** function where half the observations lie in each set. Fit a regression tree to the training set. Plot the tree. Which predictor variables were used in the regression tree?
- (b) What is the test set mean squared error? Draw a scatter plot of the fitted and true values.
- (c) Use the default settings and cross-validation in order to determine the optimal level of tree complexity. Would you prune the tree based on the result?
- (d) If you decide to prune the tree in the previous step, plot the new tree. What is the test set mean squared error? Compare with the result in part (b).
- (e) Use random forests to analyze this data. Set the seed to 1 before running the method. What test set mean squared error do you obtain? How does this compare to the CART model? How many variables does the randomForest function try at each split?

- (f) Use the **importance()** function to determine the two variables which are most important. Plot the importance measures using the **varImpPlot()**.
- (g) Describe the effect of the number of variables considered at each split controlled by the **mtry** argument in **randomForest()**, on the error obtained.